

Proof of Concept: HTTP Log Attack Analysis

Splunk-based Investigation of http logs

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1. Objective

This POC documents the investigation of the HTTP connection log (sourcetype="http logs") imported into Splunk, to identify, validate, and quantify web-based attack activity on the internal network. Two distinct attacks were identified. Each finding below is backed by the exact SPL query used, a screenshot of the result returned by Splunk, and the reasoning used to conclude whether the result confirms an attack.

2. Data Source

- HTTP log — sourcetype="http logs", Zeek HTTP connection records, imported into Splunk as raw unstructured text.

Because the file was ingested without field extraction configured, every query below begins by extracting the needed fields from _raw using rex before any analysis is performed.

3. Attack #1 — DirBuster Directory Brute-Force (192.168.203.63)

3.1 Main Evidence — Top Source by Volume

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]
+)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| stats count as events, dc(uri) as unique_uris by src_ip
| sort -events
```

The screenshot shows the Splunk search interface. The search bar contains the same SPL query as above. Below the search bar, a message states: "Events might not be returned in sub-second order due to search memory limits. See search.log for more information. Increase the value of the following limits.conf setting: [search]:max_rawsize_perchunk." The search results show 2,043,366 events. The table below displays the top sources by volume.

src_ip	events	unique_uris
192.168.203.63	1284535	1262909
192.168.202.79	232259	210554
192.168.202.102	179574	20166
192.168.202.110	169065	30627
192.168.202.138	47379	8927
192.168.202.140	28332	5871

Figure 1 — 192.168.203.63 tops the list with 1,284,535 events and 1,262,909 unique URIs — by far the largest source in the entire log, out of 2,043,366 total events.

A single source generating 1.28 million requests — over 62% of the entire log — with over a million distinct URI guesses, is not normal browsing. This is the core evidence for this attack; the sections below present the supporting evidence gathered to validate it.

3.2 Supporting Evidence

3.2.1 Tool Signature Confirmation

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]+)
)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search src_ip="192.168.203.63"
| stats count by user_agent
| sort -count
```

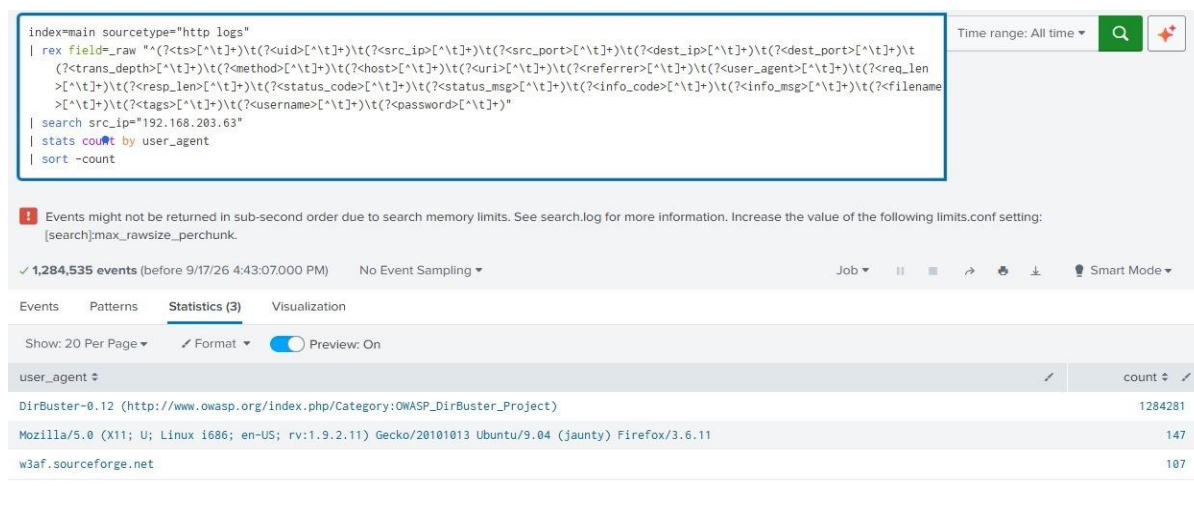


Figure 2 — Of 1,284,535 requests from this source, 1,284,281 carry the User-Agent “DirBuster-0.12 (http://www.owasp.org/index.php/Category:OWASP_DirBuster_Project)” — a real, well-known directory-brute-forcing tool.

3.2.2 Target Concentration

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]+)
)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search src_ip="192.168.203.63"
```

```
stats count by dest_ip
sort -count
```

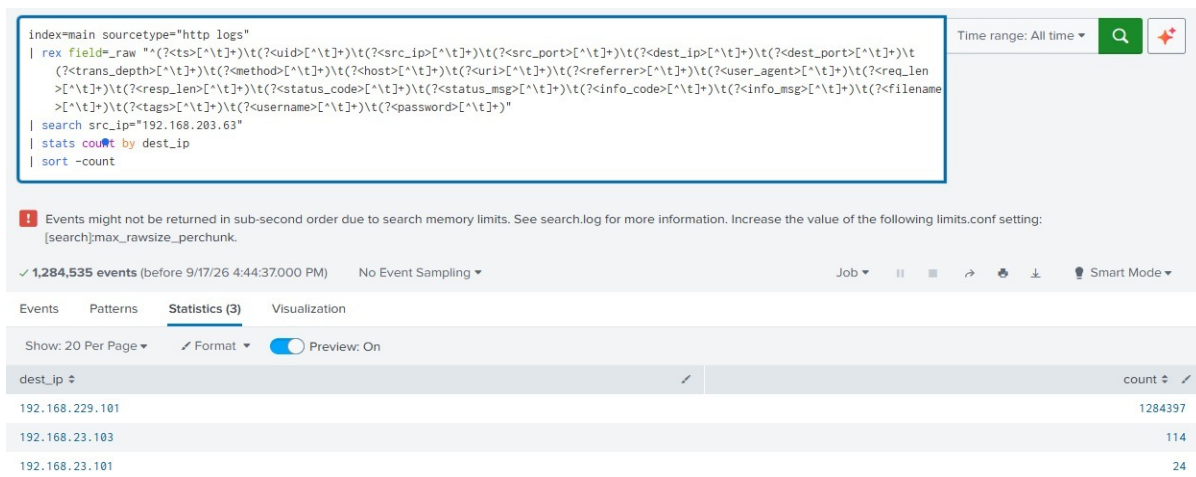


Figure 3 — 1,284,397 of this source's 1,284,535 requests — nearly all of them — were aimed at a single host, 192.168.229.101. Only 114 and 24 requests touched two other hosts.

3.2.3 Blind-Enumeration Confirmation (Status Codes)

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]
+)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search src_ip="192.168.203.63"
| stats count by status_code
| sort -count
```

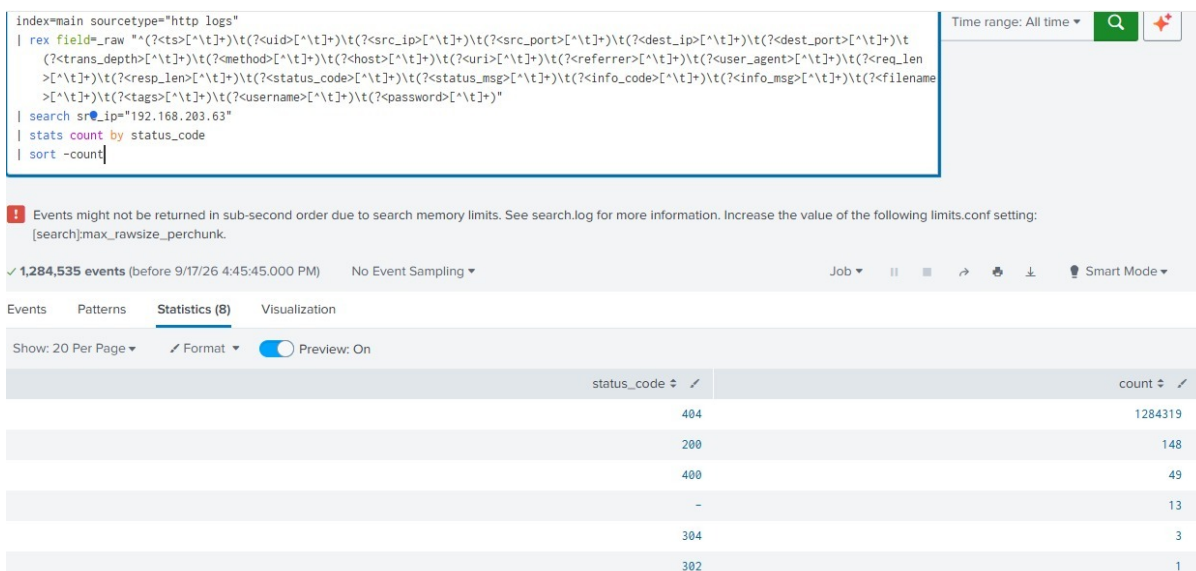


Figure 4 — 1,284,319 of 1,284,535 requests (99.98%) returned 404 Not Found. Only a handful returned 200, 400, or other codes — confirming this was blind guessing, not real browsing of existing content.

Together, the tool signature, the near-total concentration on a single host, and the near-total 404 failure rate confirm this was an automated directory brute-force scan carried out with DirBuster — not legitimate traffic.

4. Attack #2 — Web Enumeration & Multi-Vector Exploitation (192.168.202.79)

4.1 Main Evidence 1 — Enumeration Volume

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]+)
)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search src_ip="192.168.202.79"
| stats count as events, dc(dest_ip) as targets, dc(uri) as unique_uris
```

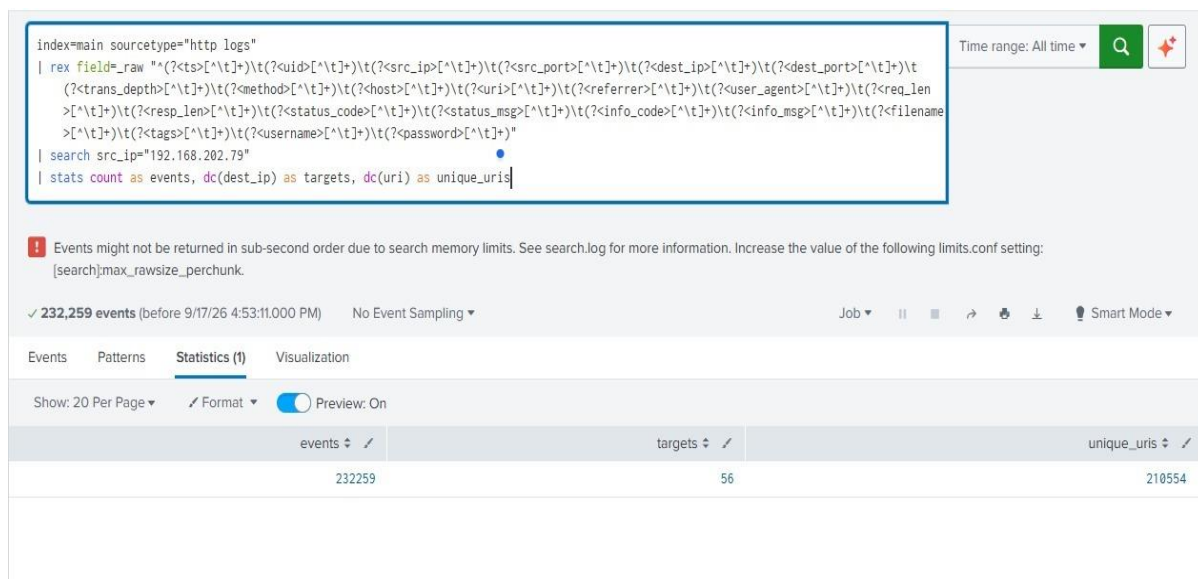


Figure 5 — 192.168.202.79 made 232,259 requests against 56 different destination hosts, using 210,554 unique URIs — a systematic enumeration sweep, not normal browsing.

4.2 Main Evidence 2 — Injection / Traversal Payloads

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]+)
)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search src_ip="192.168.202.79" (uri="*.*/*" OR uri="*etc/passwd*" OR uri="*boot.ini*"
OR uri="*vmInventory*" OR uri="*WPnBr.dll*" OR
uri="*<script>*" OR referrer="*<script>*" OR referrer="*system(*)"
| stats count
```

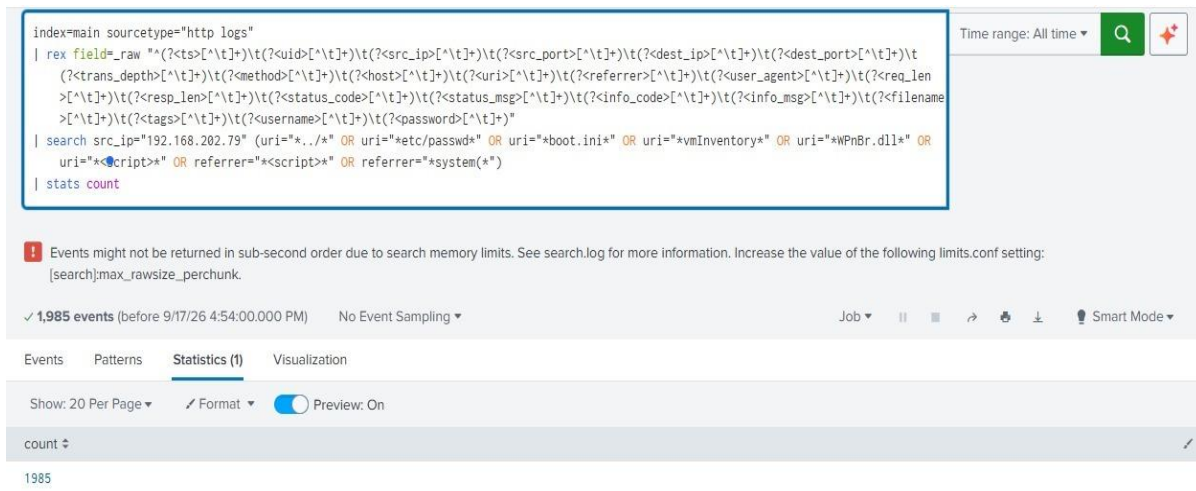


Figure 6 — 1,985 of this source's requests carried real attack payloads — directory traversal (../), attempts to read /etc/passwd and boot.ini, a VMware config exfiltration path, a known IIS exploit path (WPnBr.dll), and XSS/command-injection strings embedded directly in the URL.

4.3.1 SQL Injection Attempts (192.168.202.110)

SQL injection is an attack where malicious database commands are inserted into a normal-looking web request (a URL or form field), tricking the website's database into running them. If successful, it can let an attacker read, modify, or delete data they should never have access to — or in some cases bypass a login entirely.

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]
+)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search tags="HTTP::URI_SQLI"
| stats count by src_ip
| sort -count
```


Figure 8 — Actual request URIs from 192.168.202.110, showing real SQL injection syntax embedded directly in the URL — UNION SELECT statements and a classic 1'or'1=1 tautology — not just a tag, but the literal attack payload itself.

4.3.3 Outcome of the SQL Injection Attempts (192.168.202.110)

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]
+)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search tags="HTTP::URI_SQLI" src_ip="192.168.202.110"
| stats count by status_code
```

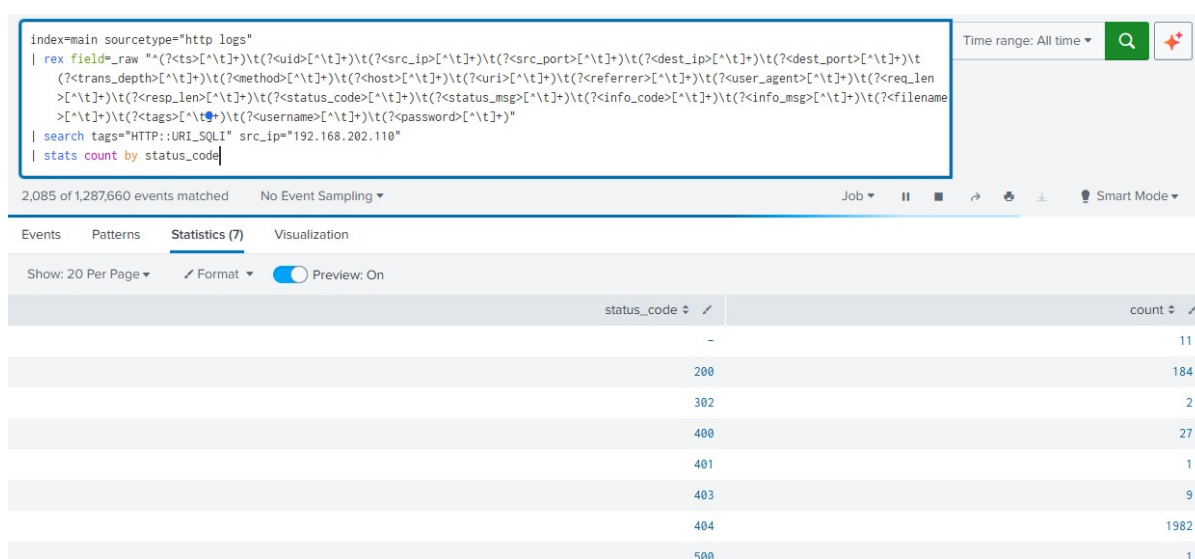


Figure 9 — Of this source's 2,085 SQL injection attempts, 1,982 returned 404 and were blocked, but 184 returned 200 OK — meaning the server processed and responded normally to those requests, and 9 returned 403 Forbidden, suggesting some form of access-control response.

4.3.4 Credential Brute-Force Against Nagios (192.168.202.96)

```
index=main sourcetype="http logs"
| rex field=_raw
"^(?<ts>[^\t]+\t(?<uid>[^\t]+\t(?<src_ip>[^\t]+\t(?<src_port>[^\t]+\t(?<dest_ip>[^\t]
+)\t(?<dest_port>[^\t]+\t
(?<trans_depth>[^\t]+\t(?<method>[^\t]+\t(?<host>[^\t]+\t(?<uri>[^\t]+\t(?<referrer>[
^\t]+\t(?<user_agent>[^\t]+\t(?<req_len
>[^\t]+\t(?<resp_len>[^\t]+\t(?<status_code>[^\t]+\t(?<status_msg>[^\t]+\t(?<info_cod
e>[^\t]+\t(?<info_msg>[^\t]+\t(?<filename
>[^\t]+\t(?<tags>[^\t]+\t(?<username>[^\t]+\t(?<password>[^\t]+)"
| search src_ip="192.168.202.96" username="guest"
| stats count by uri, dest_ip
```

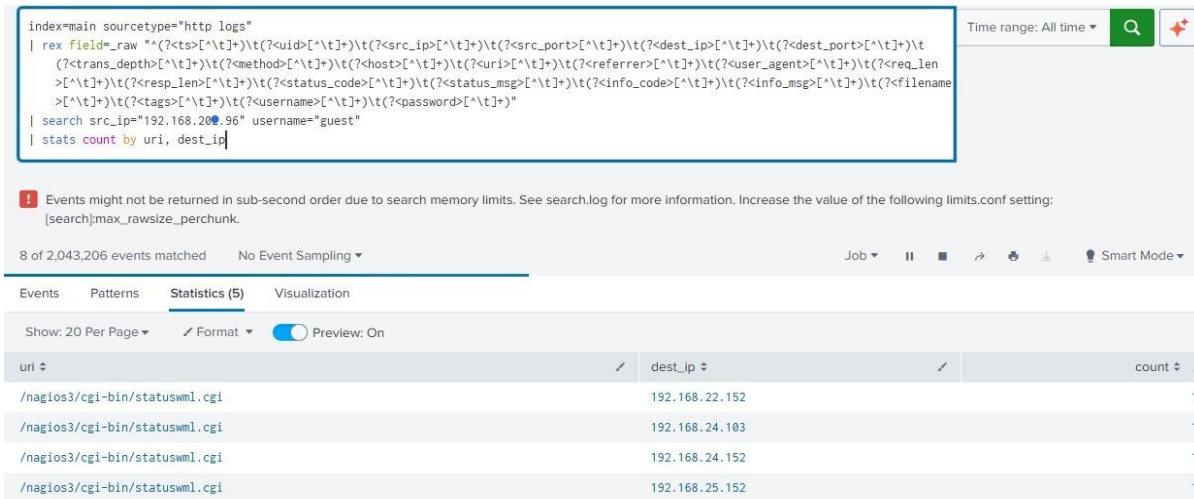


Figure 8 — 8 attempts using username “guest” were made from 192.168.202.96, all against /nagios3/cgi-bin/statuswml.cgi — a known Nagios exploit path — across 5 different destination hosts.

Taken together, the enumeration sweep and injection payloads from .79, the SQL injection attempts concentrated on .110, and the Nagios credential guessing from .96 show multiple IPs in the same incident window moving beyond reconnaissance into active exploitation attempts — reading sensitive files, injecting code and database commands, and guessing admin credentials.

5. Overall Finalization

Each attack was validated using an evidence checklist rather than a single indicator. The table below summarizes which criteria were confirmed, based only on the queries and screenshots gathered above.

Attack	Criteria	Evidence
Attack #1	High-volume automated scanning	1,284,535 requests from a single source — over 62% of the entire log
Attack #1	Tool signature confirmed	DirBuster-0.12 on 1,284,281 of 1,284,535 requests
Attack #1	Concentrated targeting	1,284,397 requests aimed at one host, 192.168.229.101
Attack #1	Blind enumeration confirmed	1,284,319 of 1,284,535 requests (99.98%) returned 404
Attack #2	Web enumeration confirmed	232,259 requests, 56 targets, 210,554 unique URIs from 192.168.202.79
Attack #2	Injection / traversal payloads	1,985 requests carrying real attack payloads (traversal, XSS, command injection)
Attack #2	SQL injection confirmed	2,827 Zeek-tagged HTTP::URI_SQLI events, 2,217 from 192.168.202.110
Attack #2	SQL injection payload confirmed	Raw URIs show real UNION SELECT and 1'or1=1 syntax, not just a tag
Attack #2	SQL injection outcome	Of 2,085 attempts from .110: 1,982 blocked (404), 184 returned 200 OK
Attack #2	Credential brute-force confirmed	8 “guest” login attempts from 192.168.202.96 against a Nagios exploit path

Final Verdict: Both attacks are confirmed by the evidence gathered. Attack #1 (192.168.203.63) is a clear, isolated, tool-signed directory brute-force scan — the single largest source of traffic in the entire log by a wide margin. Attack #2 is a separate, multi-IP campaign centered on 192.168.202.79 that moved beyond scanning into active exploitation: real injection and traversal payloads, a distinct SQL injection campaign from 192.168.202.110 with actual UNION SELECT payloads and 184 requests returning 200 OK, and credential guessing against a known Nagios exploit path from 192.168.202.96. Together these represent two distinct, confirmed attack patterns within the same incident window.